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Meta-cluster driven ensemble learning with cluster-augmented features for explainable heart disease prediction

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Heart disease remains a major global cause of death, necessitating accurate and interpretable predictive models. In this study the work proposes a Meta Cluster-Driven Ensemble Learning Engine (MCDELE) to improve predictive accuracy. Our proposed model uses a two-tier stacking architecture. The first layer consists of a range of base learners trained on a combined heart disease dataset collected from Kaggle, which is derived from multiple UCI Heart Disease cohorts. This layer is enhanced with two cluster-augmented features that represent structural patterns within the dataset. Four statistical meta-features are then computed from the base learners' predictions and passed on to the second meta-learner layer. The proposed ensemble achieved the highest accuracy (90.22%), F1-score (91.26%), and precision (90.38%) while comparing with the baseline models, and attaining a strong AUC of 93.77%. The role of cluster-augmented features and meta-features in the model's decision is emphasized by SHAP-based interpretation. The statistical analysis using the Wilcoxon signed-rank test supports the competitiveness of MCDELE over several competing models across multiple evaluation metrics. The finding demonstrates that MCDELE is a reliable and interpretable framework for a medical decision support system.

Keywords Heart disease prediction, Ensemble learning, MCDELE, Cluster-augmented feature, SHAP, Wilcoxon signed-rank test

Cardiovascular Diseases (CVDs), especially heart disease, remain a major global health concern[1]. As per the World Health Organization (WHO) estimation, 19.8 million people die in a year due to heart disease, which is 32% of all deaths worldwide[2]. In order to avoid complications and enhance patient outcomes, early and accurate detection is essential. In recent years, artificial intelligence (AI), especially machine learning (ML) and deep learning (DL), has demonstrated tremendous promise in automating and enhancing the prediction of heart disease, allowing for data-driven, scalable, and impartial healthcare decision-making. Several classical ML models like Logistic Regression (LR), Support Vector Machine (SVM), Decision Trees (DT), K-Nearest Neighbors (KNN), and Naïve Bayes (NB) are widely used in multiple studies and show strong performance on structured feature sets[3–6]. Deep learning models like Deep Neural Network (DNN), Convolutional Neural Network (CNN), and Recurrent Neural Network (RNN) are proven effective in processing unstructured data, medical images, and time series datasets[7–10].

Heart disease is difficult to diagnose early, even with major advances in cardiovascular diagnostics, because clinicians must evaluate a large volume of patient information and access multiple interacting risk factors. This can be time-consuming, and traditionally, the subjective diagnostic approaches may have variability between practitioners. Therefore, AI-based clinical decision-support systems planning to make an accurate assessment of patient risk in a consistent manner are appealing. Consequently, machine learning methods have proved to be an effective tool for supporting doctors in predicting heart disease at an early stage.

Ensemble learning techniques have gained popularity as a way to further increase the accuracy and robustness of the existing predictive models by combining the strengths of multiple individual models. Several ensemble techniques, such as Bagging (e.g., Random Forest), Boosting (e.g., Gradient Boosting, XGBoost, LightGBM),

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and Stacking, are successfully used in heart disease prediction. Bagging focuses on training multiple models on a distinct dataset to reduce the variance[11], whereas boosting focuses on improving model bias by sequentially fixing the mistakes of weak learners[12]. On the other hand, stacking uses a meta learner to combine the output of the base learners in the best possible way[13].

Interpretability of the ML models in decision-making is as important as prediction performance. Clinicians find it difficult to trust a well-performing black-box model unless their decision-making ability is transparent. Explainable AI (XAI) is introduced to overcome this issue. SHAP (Shapley Additive exPlanations) has drawn interest for its solid theoretical foundation and capacity to offer both local and global explanations[14–16]. Even with these developments, a lot of research still functions as a black box, providing little insight into the decision-making process. Furthermore, they frequently overlook the structural or relational patterns in the data, focusing solely on feature-based information. Additionally, ensemble predictions usually combine the output without meaningfully modelling the agreement, disagreement, or confidence of the base models.

To address these issues, we propose a two-layer hybrid stacking ensemble model called MCDELE by incorporating cluster-driven feature engineering and introducing structural meta-features. To enhance the generalizability and prediction accuracy, a range of base learners is used along with a meta-learner. The key novelties include:

- Cluster-Driven feature augmentation for base learners to capture localized patterns in the data distribution.
- Meta-feature integration for the meta-classifier to quantify the level of consensus and variations in base predictions.
- Comprehensive performance comparison with state-of-the-art classifiers across multiple performance metrics.
- Statistically significant test through Wilcoxon Signed-Rank tests in a 10-fold cross-validation environment with multiple evaluation metrics.
- Interpretation of the base-learner and meta-learner using SHAP to improve the model's interpretability.

Related work

Ozcan and Peker[17] applied the CART algorithm for heart disease classification and achieved 87.25% accuracy with minimal preprocessing. Saboor et al.[18] evaluated different ML models on a merged dataset, concluding that SVM gives better performance with data standardization and hyperparameter tuning. Ghasemieh et al.[13] have proposed the Stacking Ensemble Learner with six different base learners and XGBoost as the meta-learner, finding that a larger number of base learners with XGBoost as the meta-learner yields the best performance. Chang et al.[19] have used voting with random forest for heart disease classification and achieved nearly 82% accuracy. Abbas et al.[20] developed a voting-based ensemble model combining different ML and DL algorithms, using SMOTE for balancing the dataset, and RF is used for feature extraction, concludes that the proposed model outperforms the baseline models.

Cheng et al.[21] proposed an ensemble model combining NB, XGB, DT, SVM, and LR for the prediction of one-year mortality in elderly CHD patients and found that the ensemble model outperforms the baseline models. For heart disease prediction, Muhammad et al.[22] have tested several ML and ensemble models with different feature selection techniques, finding ET with relief feature selection delivered the best result. Bhatt et al.[23] integrated clustering with the ML models for the prediction of heart disease and found that k-modes clustering enhances the performance of the ML models. Natarajan et al.[24] have proposed both voting and stacking approaches, combining with feature selection, and found that voting without feature selection performs well, while stacking with feature selection gives the best performance. Chandrasekhar and Peddakrishna[25] have proposed a soft voting ensemble model and done performance evaluation on two different datasets, and concluded that the proposed soft voting performs better than the existing models.

Asif et al.[26] employed bootstrap sampling to create an ensemble model from four different base models for the prediction of heart disease, and found that the bootstrapped ensemble model gives higher accuracy when ET is used as the base model. Ganie et al.[27] compared the performance of voting and stacking on two different heart disease datasets, combining several existing models. They used XAI with SHAP and gave a detailed explanation for the finding and concluded that stacking is superior to the voting ensemble. Sreekumari et al.[28] conducted a performance analysis of several ML algorithms, including Voting, DNN, LR, NB, DT, etc., and found Voting with brute-force feature selection performs better than the other combinations. Manish et al.[29] proposed an RF-GB-SVM-based stacking ensemble model with LR as the meta learner, which beats existing models in heart disease prediction. Al-Mahdi et al.[30] proposed an ensemble deep NN combining multiple MLPs, where GA is used for feature selection and the Tunicate Swarm Algorithm (TSA) is used for optimal weighting of different ML models.

Shaikh et al.[31] proposed a stacking ensemble combining LR, SVC, MLP, and DT, using SVC as the meta learner for heart disease prediction. The authors have used an isolation forest for outlier detection and removal and SMOTE for dataset balancing, which improved the overall model performance. Zhou et al.[32] proposed a stacking ensemble with DT, XGB, LGBM, and RF as the meta learner. The author used network-based feature engineering and 3 features: node, edge, and rank, which were augmented with the dataset for the prediction task. Daza et al.[33] tested different stacking combinations for heart disease prediction and found that stacking with LR as a meta model performs best. Mienye et al.[34] applied Bayesian optimization for hyperparameter tuning of different ensemble models like RF, AB, and XGB, concluding that the approach enhances the performance of the model, and XGB gives the best performance. Shah et al.[35] proposed a stacking algorithm combining several base learners and XGB as the meta learner. The researchers have also computed some meta-features and used them to enhance the performance of the meta-learner. Recent research has shown the utilization of advanced ensemble and deep-learning architectures that are effective in disease diagnosis. Al-Tam et al.[36] proposed a hybrid framework to

recognize cardiovascular disease through phonocardiography (PCG) heart sound recordings, consisting of a Transformer Encoder conventionally fine-tuned with a convolutional residual. This strategy contains attention and deep feature extraction in order to enhance audio signals containing cardiovascular anomalies. On the other hand, this framework is for heart sound analysis and requires signal-based input rather than structured clinical attributes. Similarly, Al-Tam et al.[37] used a stacking ensemble learning approach for the diagnosis of Parkinson's disease by combining a number of machine-learning classifiers together by constructing a hierarchical ensemble architecture. Their results showed that stacking-based methods can improve diagnostic capability in medical applications.

The literature shows substantial progress in heart disease prediction using machine learning, ensemble learning, and deep learning techniques. Classical models like CART, SVM, and LR have competitive performance, whereas ensemble approaches, especially voting and stacking, are generally able to beat individual classifiers. Among these, some works have investigated feature selection, hyperparameter optimization, evolutionary algorithms, and resampling methods to solve problems of class imbalance and make true generalization. Nonetheless, most current approaches mainly improve predictive accuracy, largely ignoring latent data structure, ensemble diversity, and model interpretability. Most stacking algorithms rely solely on the outputs of base learners during meta-learning and do not explicitly model prediction consensus, disagreement, or confidence. Furthermore, feature augmentation based on clustering has been underutilized for heart disease prediction, although it can capture localized structure in heterogeneous patient populations. SHAP and many other XAI techniques are increasingly further enhanced, but still primarily used as a post-hoc analysis tool instead of being integrated into the transparent clinical decision support framework. These limitations inspire the design of MCDELE, a stacking ensemble that incorporates cluster-driven feature augmentation, diversity-aware meta-feature engineering, and SHAP-based interpretability within one unified framework.

Proposed model

The proposed Meta Clustering-Driven Ensemble Learning Engine (MCDELE) is a multistage pipeline intended to enhance heart disease prediction by combining feature engineering, ensemble learning, and model explainability. The process begins with data collection and preprocessing, including cleaning, encoding, and scaling of the dataset on heart disease. To enhance the data representation, cluster-augmented features are derived using the k-Means clustering algorithm and appended to the dataset. This enriched dataset is used to train a diverse set of base learners – Logistic Regression (LR), Random Forest (RF), Gradient Boosting (GB), XGBoost (XGB), and Extra Tree Classifier (ET). The prediction probability of the base learners and statistical meta-features computed from the prediction probability is used to generate the meta-feature vector, which is fed to a Logistic Regression (LR) meta-learner to produce the final prediction, following a standard stacking approach[38]. SHAP is used for both global and local explainability to ensure the reliability and transparency of the proposed model. The overall architecture is demonstrated in Fig. 1.

Dataset

In this study, we use the Heart Failure Prediction dataset collected from Kaggle[39]. This dataset consists of 918 patient records and 11 clinical attributes, with a binary target variable where 0 means heart disease is absent and 1 means heart disease is present. According to the dataset repository, this is an extended and more highly consolidated replication of the UCI Heart Disease (UCI) dataset[40], combining records from 4 databases - Cleveland, Hungary, Switzerland, and VA Long Beach. This unified dataset not only provides a larger and more diverse sample but also retains the original clinical characteristics and diagnostic variables used for heart disease prediction. Before model development, we explored the data for missing values, inconsistencies in entries, and duplicate records. If duplicate instances were discovered using all the feature values, they were removed to avoid bias in the training and evaluation processes. One-hot encoding is used to transform categorical features such as sex, chestPainType, restingECG, ExerciseAngina, and ST_slope into a numerical format. All the numerical features are then normalized using a standard scaler to bring them into a consistent range, improving the convergence and performance. The dataset has 508 (55.34%) positive instances for heart disease and 410 (44.66%) as negative, resulting in a moderate class imbalance binary classification problem. Table 2 represents a more detailed view of the dataset and its feature descriptions. The dataset was split using a stratified 80:20 train-test split, where 80% of the data were used for model development and internal validation, whereas the other 20% were kept as an independent holdout test set for model comparison and statistical analysis.

Data preprocessing

Standard processing steps are used to ensure the model's fairness and dependability. Data standardization and balancing are optionally (based on hyper-parameter) applied to the data if the data is not already standardized or balanced before passing to the model. Let $D = \{(x_i, y_i)\}_{i=1}^{N}$, $x_i \in R^d$, $y_i \in \{0,1\}$is the training set on N samples and binary labels. A standard scaler $\tilde{x_i} = S(x_i)$, S: $R^d \to R^d$ is applied so that $\tilde{x_i}$ has zero mean and unit variance per feature. SMOTE-Tomek is a hybrid approach that first creates synthetic minority samples using SMOTE[41] and then eliminates borderline or overlapping samples using Tomek-links[42] to generate a more balanced and cleaner dataset[43]. SMOTE-Tomek is used not only to handle the class imbalance problem but also to enhance the class separation, which is crucial for downstream classifiers and clustering models' performance.

$$\left\{\left(\tilde{x}_i',\ y_i'\right)\right\}_{i=1}^{N'} = SMOTETomek\left(\{\tilde{x}_i,\ y_i\}\right) \tag{1}$$

References	Year	Dataset(s)	Method	Resampling	Feature engineering	Interpretability	Observation
Ganie et al.[27]	2025	HDDC, UHDD	LR, RF, ET, LGBM, XGB, CB	NA	NA	SHAP	Stacking performs better than voting.
Sreekumari et al.[28]	2025	Cardiovascular Disease dataset	Voting, LR, NB, kNN, DT, NN, DNN	NA	Brute Force feature selection	NA	Voting with the feature selection technique performs best.
Manish et al.[29]	2025	Heart Statlog Cleveland Hungary dataset	Stacking, LR, RF, GB, SVM	NA	NA	NA	Stacking with hyper-parameter tuning beats the existing model.
Al-Mahdi et al.[30]	2025	UCI Cleveland Heart Disease dataset	Weighted Average, MLP	NA	Genetic Algorithm for feature selection	NA	Ensemble MLP performs better than a single DNN.
Shah et al.[35]	2025	3 publicly available heart disease datasets from IEEE Dataport	Stacking, CB, LGBM, XGB, NN, RF, SVM	SMOTE	Custom feature engineering	SHAP	Meta features are used for the performance enhancement of the meta learner.
Cheng et al.[21]	2024	Department of Geriatric Cardiology	NB, XGB, DT, SVM, LR	NA	NA	SHAP	The ensemble model outperforms the baseline models.
Natarajan et al.[24]	2024	Z-Alizadeh Sani dataset	Stacking with LR, RF, SVM	NA	Firefly for feature selection	NA	Stacking with feature selection performs better.
Daza et al.[33]	2024	Heart Failure Prediction Dataset	Stacking, DT, GB, KNN, RF, LR, NG, AB, XGB	Oversampling	NA	NA	Stacking with LR as the meta model performs best.
Mienye et al.[34]	2024	Cleveland Heart Disease dataset	RF, AB, XGB	NA	NA	SHAP	Models tuned with Bayesian optimization perform better than the regular models.
Ozcan et al.[17]	2023	IEEEDataPort Heart Disease Dataset	CART	NA	NA	Tree-based interpretation	Straightforward CART application lacks a novel methodology.
Ghasemieh et al.[13]	2023	MIT Laboratory for Computational Physiology	Stacking with LR, SVM, DT, KNN, RF, XGB	NA	NA	NA	More base models enhance the overall performance.
Abbas et al.[20]	2023	Framingham Heart Study dataset	Voting, NB, DT, KNN, RF, XGB, BNN, FT-DNN	SMOTE	RF for feature extraction	NA	The proposed model beats the baseline models.
Bhatt et al.[23]	2023	Cardiovascular Disease dataset	MLP, DT, RF, XGB	NA	K-modes with Huang	NA	K-Modes clustering enhances the performance of ML models.
Chandrasekhar et al.[25]	2023	Cleveland and the IEEE Dataport dataset	Voting, RF, KNN, LR, NB, GB, AB	NA	NA	NA	Soft voting performs better than hard voting.
Asif et al.[26]	2023	Heart Disease Cleveland UCI, Heart Disease Dataset, Heart Disease	RF, XGB, ET, CB	Bootstrap sampling	NA	NA	ET beats other models with bootstrap sampling.
Yasmeen Shaikh et al.[31]	2023	UCI Repository	Stacking, LR, SVC, MLP, DT	SMOTETomek	NA	NA	Isolation forest and SMOTE combining improve the model's performance.
Zhou et al.[32]	2023	Hospital Data from Sichuan Province	Stacking, DT, XGB, LGBM, RF	NA	Network-based feature extraction technique	SHAP	Network features are used along with selected basic information for model training.
Saboor et al.[18]	2022	Cleveland Clinic Foundation, StatLog datasets, Z-Alizadeh Sani dataset	AB, MNB, SVM, LDA, RF, ET, XGB, LR, CART,	NA	NA	NA	Data standardization and hyperparameter tuning improve the model performance.
Chang et al.[19]	2022	Heart Disease Dataset	Voting with RF	NA	NA	NA	Straightforward Voting application with RF, Lakes' novel methodology.
Muhammad et al.[22]	2020	Cleveland and Hungarian dataset	KNN, NB, SVM, DT, ET, GB, AB, RF, LR, ANN		FCBF, mRMR, LASSO, Relif	NA	ET with Relief feature selection gives the best performance.
Al-Tam et al.[36]	2024	PCG Hart Sound Recording	VGG16, ResNet50, Transformer Encoder	Yes	Feature extraction using CNN and Transformer	NA	Demonstrates that combining convolutional and attention mechanisms improves cardiovascular disease recognition from heart sound signals.
Al-Tam et al.[37]	2024	Benchmark Parkinson's disease datasets	RF, DT, LR, GB, SVM, Bagging and Stacking	SMOTE	VM and Gradient Boosting were used as feature extraction	NA	Demonstrating the benefit of combining heterogeneous classifiers
MCDELE (proposed)	2026	Heart Failure Prediction dataset	Stacking, LR, RF, GB, XGB, ET	SMOTETomek	Cluster-Augmented feature, statistical meta-feature	SHAP	Engineered features play a crucial role in decision-making.

Table 1. Literature review.

Cluster-augmented feature

To add structural awareness to the data, MCDELE uses K-Means as a clustering mechanism. K-Means is an unsupervised clustering algorithm that iteratively minimizes the intra-cluster distance between data points and their designated cluster centroids in order to partition the data into k-discrete, non-overlapping clusters[44]. Silhouette score is a popular internal evaluation metric that measures cluster cohesion and separation[45]. In our

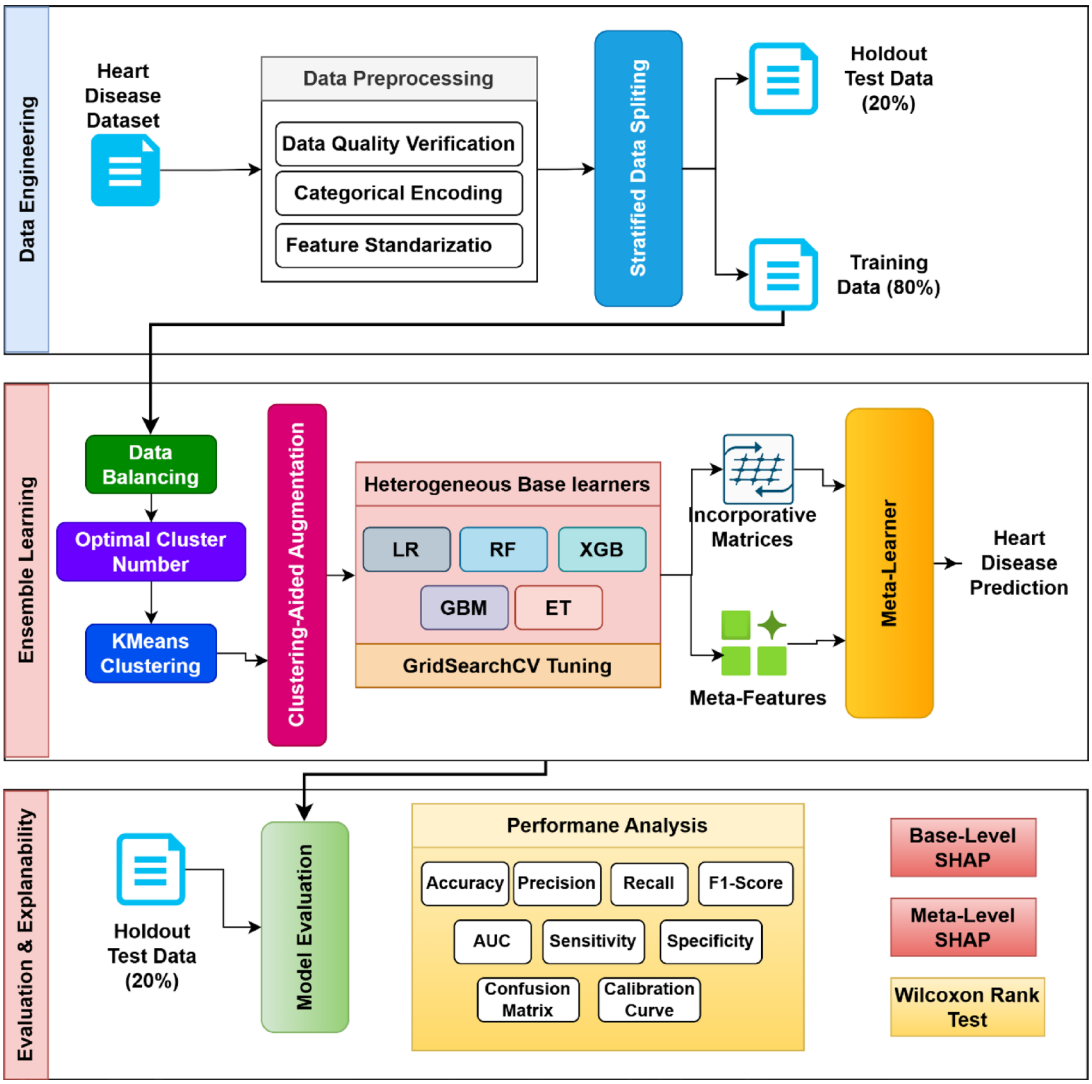


Fig. 1. MCDELE's architecture.

Feature	Description	Type	Possible values/range
Age	Age of the patient in years	Numerical	28–77 years
Sex	Biological sex of the patient	Categorical	M (Male), F (Female)
ChestPainType	Type of chest pain experienced by the patient	Categorical	TA (Typical Angina), ATA (Atypical Angina), NAP (Non-Anginal Pain), ASY (Asymptomatic)
RestingBP	Resting blood pressure measured in mm Hg	Numerical	0–200 mm Hg
Cholesterol	Serum cholesterol concentration	Numerical	0–603 mg/dL
FastingBS	Fasting blood sugar level greater than 120 mg/dL	Binary Categorical	0 = False, 1 = True
RestingECG	Resting electrocardiogram result	Categorical	Normal, ST, LVH
MaxHR	Maximum heart rate achieved during exercise	Numerical	60–202 bpm
ExerciseAngina	Exercise-induced angina	Binary Categorical	Y = Yes, N = No
Oldpeak	ST-segment depression induced by exercise relative to rest	Numerical	− 2.6 to 6.2
ST_Slope	Slope of the peak exercise ST segment	Categorical	Up, Flat, Down
HeartDisease	Presence of heart disease (target variable)	Binary Categorical	0 = No Heart Disease, 1 = Heart Disease

Table 2. Feature description table.

Input	- Labelled Dataset D = (X_train, y_train) - cluster_range = [k_{min}, k_{max}] - balance_flag ∈ {true, false} - random_seed
Output	- Trained base learners B_trained - Trained meta-learner M_meta - Fitted clustering model C_kmeans

Procedure

1. Base learners: B = {b_1, b_2, …, b_n}
2. Meta Learner: M_meta
3. if balance_flag:
   (X_bal, y_bal) ← SMOTETomek(X_train, y_train)
4. else:
   (X_bal, y_bal) ← (X_train, y_train
5. for k in cluster_range:
   Fit KMeans(k) on X_bal
   Compute silhouette_score(k)
6. Select k* = arg max_k silhouette_score(k)
7. C_kmeans, centroids ← KMeans(k*)
8. Z = []
9. for x in X_bal:
   c = C_kmeans.predict(x)
   centroid_c = C_kmeans.cluster_centers_[c]
   conf = 1 / (1 + distance (x, centroids_c))
   Z.append(augment (x, c, conf))
10. P_{OOF} = matrix(len(Z), len(B))
11. B_trained = []
12. For b_j in B:
    grid = GridSearchCV.fit(b_j, param_grid, Z, y_bal)
    best_b_j = grid.best_estimator_
    Divide (Z, y_bal) into K_{fold} stratified folds
    For each fold f ∈ {1 ... K_{fold}}:
      Train b_j (using best_b_j parameters) on all folds except f
      Predict probabilities for fold f: P_{OOF}[f][i] = b_j.predict_proba(Z[f])[:,1]
    Fit b_j (using best_b_j) on the entire Z and y_bal
    B_trained.append(b_j)
13. M = []
14. For probs in rows (P_{OOF}):
    stats = [mean (probs), std (probs), min (probs), max (probs)]
    M.append(concate(probs, stats)
15. M_meta.fit (M, y_bal)
16. Return centroids, B_ trained, M_meta, C_kmeans

Algorithm 1. MCDELE training.

Input	- Unseen Test dataset X_test - Trained base learners B_trained = {b_1, b_2, …, b_n} - Trained meta-learner M_meta - Fitted clustering model C_kmeans
Output	- Final prediction probabilities: y_MCDELE

Procedure

1. Z_test = []
2. For x in X_test:
   c = C_kmeans.predict(x)
   centroid_c = C_kmeans.cluster_center_[c]
   conf = 1 / distance (x, centroids_c)
   Z_test.append(augment(x, c, conf)
3. P_test = matrix(len(Z_test, len(B_trained))
4. For b_j in B_Trained:
   For each sample i:
   P_test[i][j] = b_j.predict_proba(Z_test[i])[:, 1]
5. M_test = []
6. For each row I in P_test:
   probs = P_test[i]
   stats = [mean (probs), std (probs), min (probs), max (probs)]
   M_test.append(concate(probs, stats)
7. Y_MCDELE = M_meta.predict_proba(M_test)[:, 1]
8. Return y_MCDELE

Algorithm 2. MCDELE prediction.

experiment, instead of using a fixed number of clusters, the optimal cluster number is chosen from a range of values [k_min, k_max] by using the Silhouette score.

Then the K-Means clustering is applied on the scaled and sampled data with the optimal cluster number:

$$\{\mu_k\}_{k=1}^{K}, argmin_{\{\mu_k\}} \sum_{i=1}^{N'} \min || \tilde{x}_i' - \mu_k ||^2 \tag{2}$$

Each $\tilde{x}_i'$ is assigned with a cluster label (c_i) and cluster confidence (k_i):

$$c_i = \text{argmin} || \tilde{x}_i' - \mu_k ||^2 \in \{1,2,\ldots,K\} \tag{3}$$

$$d_{ik} = \left|\left| \tilde{x}_i' - \mu_k \right|\right| \tag{4}$$

$$k_i = \frac{1}{1 + d_i c_i} \in (0, 1] \tag{5}$$

The confidence score is a continuous strength indicator of cluster assignment. Since K-means assigns samples to its closest centroid, distance is a natural measure of representability, which reflects how representative the sample is for its assigned cluster. Higher confidence is assigned to samples that are closer to the centroid and lower for samples located near cluster boundaries. We chose the inverse-distance transformation for its properties of returning values in, preserving monotonicity with respect to distance, and avoiding any additional hyperparameters.

Both cluster label and cluster confidence are then augmented to each sample vector:

$$z_i = [\tilde{x}_i'; e(c_i); k_i] \tag{6}$$

Where $e(c_i)$ is the one-hot encoding of the cluster label.

Base learner ensemble

A wide range of ML models is chosen for this stacking ensemble. For binary classification, Logistic Regression (LR) is a simple yet powerful linear model that offers interpretability and a solid baseline[46]. Random Forest (RF) increases the robustness as it has the ability to handle non-linear relationships and capture complex interactions[47]. Gradient Boost (GB) provides strong performance in many cases by concentrating on the errors of the previous trees[48]. XGBoost (XGB) provides high predictive power and enhances the ensemble performance by utilizing gradient boosting and regularization techniques[49]. Lastly, the Extra Trees Classifier (ET), less prone to overfitting, adds another level of diversity while preserving the high accuracy[50]. To ensure the best version of each base learner is used, GridSearchCV with a 5-fold internal stratified cross-validation strategy is used to determine the ideal hyperparameters. For LR, the regularization strength (C∈{0.01,0.1,1,10}) parameter was optimized to 0.1, with a penalty of 'l2' and the solver set to 'lbfgs'. For the boosting models (XGB and GB), the search space covered the number of estimators (50 to 200), learning rates (0.01 to 0.1), and maximum tree depths (3 to 5). The final XGB model has 100 estimators, a max depth of 3, a learning rate of 0.1, and one subsample. The GB model has 50 estimators, a learning rate of 0.1, and a max depth of 3. For RF and ET, the number of estimators is 200, max depth is unbounded, minimum samples for split and leaf node are 2 and 1, and ET uses 'sqrt' for the max_features parameter. The KMeans clustering was initialized with the k-means + + strategy with random_state = 42, and the optimal number of clusters was determined by evaluating the silhouette coefficient over a range of k = 2–10. The remaining parameters followed the default values of the implementation in scikit-learn. An out-of-fold (OOF) strategy is used to provide a valid stacking architecture and prevent information leakage during the training of the meta-learner. In particular, for every base learner, we produce predictions while running K-fold cross-validation on the training set. This guarantees that the meta-learner inputs consist of samples not seen by the base learners in their respective training cycles, providing a robust estimate of base learner performance and thwarting overfitting to in-sample biases. Algorithm 1 shows the complete training process.

Let B = {b_1, b_2, …, b_M}be the set of M base classifiers tuned with GridSearchCV and trained on $\{z_i, y_i'\}$ and the prediction probability is computed as:

$$p_{ij} = b_j\left(z_i\right) = \Pr\left(y = 1 \middle| z_i; b_j\right) \middle| \Pr\left(y = 0 \middle| z_i; b_j\right) \tag{7}$$

All the prediction probabilities are then collected into a base-prediction matrix:

$$P = \left[p_{ij}\right]_{i=1,\ldots,N',j=1,\ldots,M} \in [0,1]^{N' \times M} \tag{8}$$

Meta-feature construction

With the obtained predictions from the tuned base learners, we engineer a set of diversity-aware meta features to improve the meta learner's learning ability. From the prediction probabilities, we compute the mean: captures agreement, standard deviation: shows the variability of the predictions, minimum and maximum possible outliers, and ranges. These characteristics help the meta-learner to comprehend confidence and ambiguity in an ensemble decision by offering crucial insights. From each row $P_{i,*}$, four diversity meta-features (mean, std, min, max) are derived as follows:

$$\mu_i^{(mean)} = \frac{1}{M}\sum_{j=1}^{M} p_{ij} \tag{9}$$

$$\sigma_i^{(std)} = \sqrt{\frac{1}{M}\sum\nolimits_{j=1}^{M}\left(p_{ij} - \mu_i^{(mean)}\right)^2} \tag{10}$$

$$\mu_i^{(min)} = \min p_{ij} \tag{11}$$

$$\mu_i^{(max)} = \max p_{ij} \tag{12}$$

Then the meta-feature vector is computed as:

$$m_i = \left[P_{i,*}; \mu_i^{(mean)}; \sigma_i^{(std)}; \mu_i^{(min)}; \mu_i^{(max)}\right] \tag{13}$$

Meta-learner

Logistic Regression (LR) is selected as the meta-learner in the proposed model due to its ease of use, interpretability, and efficiency. Base learners like RF, GB, XGB, and ET often have complex non-linear decision boundaries. LR is a linear model that balances the base models' predictions by utilizing their strengths while offsetting their weaknesses[51]. Because it trains quickly, learning can be done effectively without overfitting to noise. Also, its probabilistic nature aids in the model's interpretability. So, a meta-classifier g (LR) is trained on $\{(m_i, y_i')\}_{i=1}^{N'}$ while minimizing the log-loss:

$$\hat{g} = argmin\left\{-\sum_{i=1}^{N'}\left[y_i' \log g\left(m_i\right) + \left(1 - y_i'\right)\log\left(1 - g\left(m_i\right)\right)\right]\right\} \tag{14}$$

The final prediction probability for a new sample x is:

$$\widehat{p}(x) = \widehat{g}([P_*(x); \mu^{(mean)}(x); \sigma^{(std)}(x); \mu^{(min)}(x); \mu^{(max)}(x)]) \quad (15)$$

Model evaluation

Environmental setup

The workstation used for our experiments was an Intel Core i7, 16 GB RAM with 8 GB shared Intel Iris graphics running Windows 11 in an Anaconda environment, Python 3.12 for all experiments. We implemented the proposed MCDELE framework using Scikit-learn 1.5.1, XGBoost 3.0.2, Imbalanced-learn 0.12.3, NumPy version 1.26.4, Pandas version 2.2.2, and SHAP version 0.48.0. No GPU acceleration was used while the model was developed or processed.

Evaluation metrics

The performance of MCDELE and the performance comparison with the existing models are evaluated using the following widely used classification metrics:

- Accuracy: Indicate the total percentage of instances that are correctly classified.
- Precision: Calculates the ratio of true positive predictions to all positive predictions, indicating the model's reliability in identifying the positive classes.
- Recall: Indicates the ability to accurately recognize all true positive instances.
- F1-Score: A single metric that balances precision and recall by taking the harmonic mean.
- Area Under ROC Curve (AUC): Indicate the model's ability to differentiate between positive and negative classes, especially in imbalanced datasets.

These metrics represent a comprehensive view of the model's performance, completeness, and decision confidence.

Statistical significance testing

The Wilcoxon Signed-Rank test is used in this study to evaluate the proposed model's performance. This non-parametric test compares performance metrics of two models without assuming a specific data distribution. This approach is especially helpful when the sample size is small or the data has a non-normal distribution[52]. The statistical analysis was conducted separately from the hold-out evaluation, using a 10-fold cross-validation procedure on the complete dataset; the models' accuracy, f1-score, and AUC are measured. The significance level of $\alpha = 0.05$ is used to determine whether the observed difference is statistically significant or not.

Results and discussion

Table 3 shows the summary of 10-fold cross-validation results and confirms that the proposed model is robust, with a mean accuracy of 90.04% and a mean AUC of 92.59%. These results confirm that the framework maintains good predictive performance with low variability, which is reflected in the small standard deviations for each metric we record. Table 4 presents the benchmarking results of the proposed MCDELE model against a set of classifiers evaluated on the independent 20% hold-out test set. The baseline models include Logistic Regression (LR), Support Vector Machine (SVM), AdaBoost (AB), Extra Trees (ET), Gradient Boosting (GB), Random Forest (RF), XGBoost (XGB), LightGBM (LGBM), CatBoost (CB), Balanced Random Forest (BRF), and Easy Ensemble (EE). The comparative evaluation is conducted across five key metrics: accuracy, precision, recall, F1-score, and AUC to ensure the fair assessment of model effectiveness. The proposed MCDELE shows a superior performance than the existing model on the majority of the evaluation metrics, with the highest accuracy (90.22%), precision (90.38%), and F1-score (91.29%). It shows a competitive AUC (93.77%) only slightly behind ET. This shows the MCDELE's reliability in differentiating between positive and negative classes. The model's

Fold	Accuracy	Precision	Recall	F1-Score	AUC
0	92.39	92.31	94.12	93.21	96.7
1	91.13	93.62	88.27	90.87	94.69
2	89.7	89.13	88.39	88.76	91.06
3	86.96	85.45	92.16	88.68	93.64
4	88.61	87.02	86.27	86.64	87.14
5	88.04	87.04	92.16	89.53	92.11
6	89.13	90.2	90.2	90.2	92.64
7	90.13	89.68	92.16	90.9	94.5
8	91.52	90.02	91	90.51	90.98
9	92.81	90.76	96	93.31	92.39
Mean	90.042	89.523	91.073	90.26	92.585
Std. Dev.	±1.91	±2.49	±2.91	±2.03	±2.60

Table 3. 10-fold cross-validation performance of MCDELE.

Model name	Accuracy	Precision	Recall	F1	AUC
LR	88.59	87.16	91.14	90.05	93.05
SVM	71.74	75	73.53	74.26	79.79
AdaBoost	89.13	89.2	90.2	90.2	91.73
ExtraTrees	89.67	89.52	90.16	90.82	94.67
GradientBoost	87.5	89.11	88.24	88.67	93.21
RandomForest	89.67	89.52	90.16	90.82	93.24
XGBoost	86.41	87.38	88.24	87.8	92.26
LightGBM	85.87	86.54	88.24	87.38	92.34
CatBoost	89.67	88.92	89.22	90.55	93.75
BalancedRF	88.04	89.22	89.22	89.22	93.81
EasyEnsemble	88.04	90	88.24	89.11	91.48
MCDELE	**90.22**	**90.38**	**92.16**	**91.26**	**93.77**

Table 4. Performance comparison table between MCDELE and existing ML models.

Model	Accuracy	Precision	Recall	F1	AUC
Baseline Stack	83.13	81	82.22	81.61	83.96
Stacking + SMOTE-Tomek	85.67	86.52	88.16	87.33	83.45
Stacking + SMOTE-Tomek + Cluster Info	87.13	88	89.22	88.61	83.76
Stacking + SMOTE-Tomek + Meta Info	87.76	89.48	90.14	89.81	90.99
MCDELE	**90.22**	**90.38**	**92.16**	**91.26**	**93.77**

Table 5. Ablation study results showing the impact of SMOTE-Tomek balancing, cluster information, and meta-information on the performance of the stacking ensemble.

balance of recall and precision makes it suitable in healthcare, where false positives and false negatives can be expensive.

An ablation study is also performed to assess the importance of each of the proposed components by incrementally adding SMOTE-Tomek balancing, cluster information, and meta-information starting from a baseline stacking ensemble. Both cluster information and meta-information pushed the baseline stack in the right direction, as shown in Table 5. But meta-information together produced significantly more gain than cluster information alone. The best performance on all metrics was recorded by the full MCDELE model, proving that cluster-aware learning and diversity-aware stacking not only can be effectively combined but also serve to complement each other in improving predictive accuracy.

According to the ROC comparison in Fig. 2, MCDELE matches or outperforms the majority of the standard classifiers. The performance of MCDELE shows how cluster-aware ensemble learning and meta-feature integration can enhance classification performance across changeable thresholds.

Calibration and classification performance

Calibration and classification analyses were conducted on data from the independent holdout test set to assess the validity and reliability of the MCDELE model. Figure 3 shows the confusion matrix of our test data. Overall, the model correctly classified 166 of 184 patients for an accuracy of 90.2%. Obtained 94 true-positive predictions and 72 true-negative predictions, and observed only eight false-negative and ten false-positive predictions, respectively. The resulting sensitivity and specificity were 92.2% and 87.8%, respectively.

The calibration curve of the proposed model is illustrated in Fig. 4. The calibration curve followed the ideal reference line well, indicating good agreement between predicted probabilities and observed outcomes. In the intermediate probability range, only small differences were observed, indicating a mild tendency toward conservative risk estimation. In general, the results indicate the MCDELE model has good discrimination ability and probability calibration.

Performance across demographic subgroups

To examine the robustness of the proposed MCDELE model across demographic groups, subgroup analyses (by sex and age categories) on the independent holdout test set were undertaken (Table 6). Model performance for male participants ($n = 146$): AUC: 0.914, sensitivity: 92.7%, specificity: 82.0%. In the female subset of participants ($n = 38$), an AUC value of 0.990, with a sensitivity range 83.3% and a specificity range 96.9%. Analyses by age showed high performance across all ages. Each discrimination was maximally seen in participants younger than 50 years (AUC = 0.951), followed by 50–60 years old (AUC = 0.921) and > 60 years old (AUC = 0.886). The sensitivity was consistently high across all age groups (88.5% to 96.4%) and indicates that heart disease cases were reliably detected. The subgroup analysis is descriptive and no formal statistical comparisons between

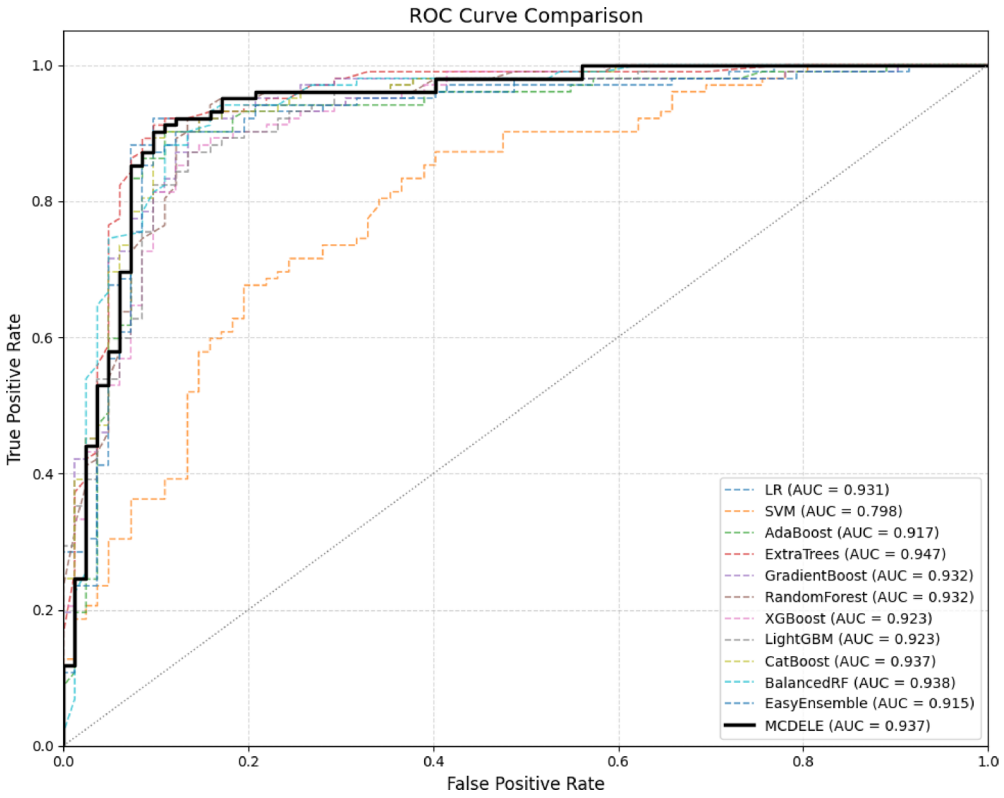


Fig. 2. ROC curve comparison between MCDELE and existing ML models.

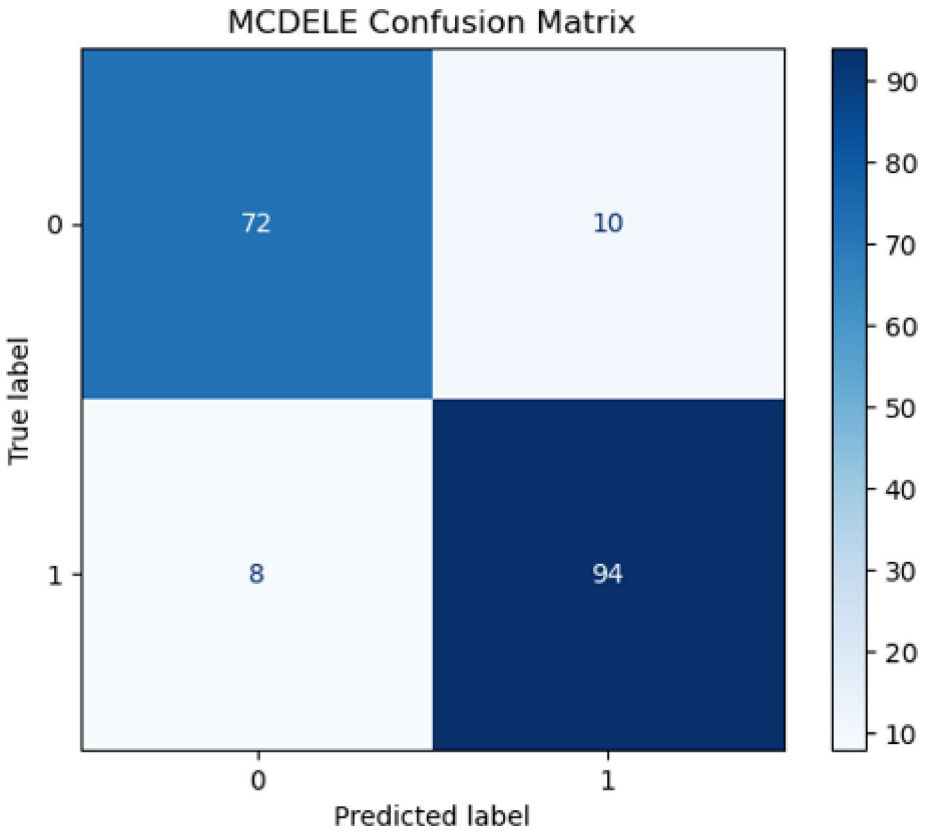


Fig. 3. Confusion Matrix of MCDELE on the holdout test set.

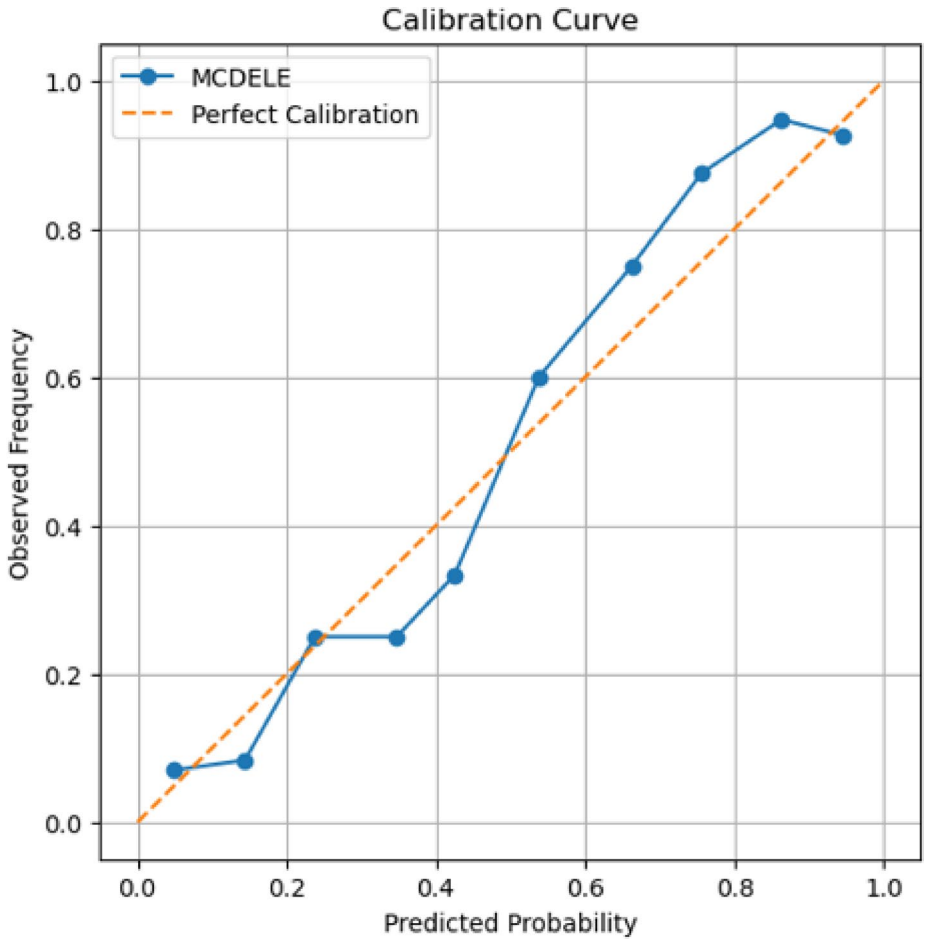


Fig. 4. Calibration Curve of MCDELE on the holdout test set.

Group	*N*	AUC	Sensitivity	Specificity
Overall	184	0.9377	0.9216	0.878
Sex = M	146	0.9138	0.9271	0.82
Sex = F	38	0.9896	0.8333	0.9688
Age < 50	68	0.9505	0.8846	0.9048
Age 50–60	83	0.9208	0.9167	0.8571
Age > 60	33	0.8857	0.9643	0.8

Table 6. Subgroup analysis of MCDELE on holdout test set.

subgroups were made. Thus, the results, especially for the smaller subgroup of females and the age group > 60 years, should be interpreted with appropriate caution due to limited sample sizes.

Statistical significance

The differences in performance were compared between MCDELE and the baseline models using the Wilcoxon Signed-Rank test with Benjamini–Hochberg (FDR) correction. Figure 5 illustrates that a majority of the baseline models have adjusted p-values below the significance threshold ($\alpha = 0.05$) for at least one evaluation metric, demonstrating that MCDELE outperforms these baseline models significantly. More specifically, we show statistically significant improvements against AdaBoost, BalancedRF, Gradient Boosting, and even SVM, across one or more performance measures. On the other hand, as there was no statistically significant difference between MCDELE and EasyEnsemble, CatBoost, and ExtraTrees upon FDR correction, these ensemble-based methods performed equally well as MCDELE on a majority of metrics. However, the MCDELE always obtained better mean results in terms of accuracy, F1-score, and AUC, which indicates that our proposed framework is effective.

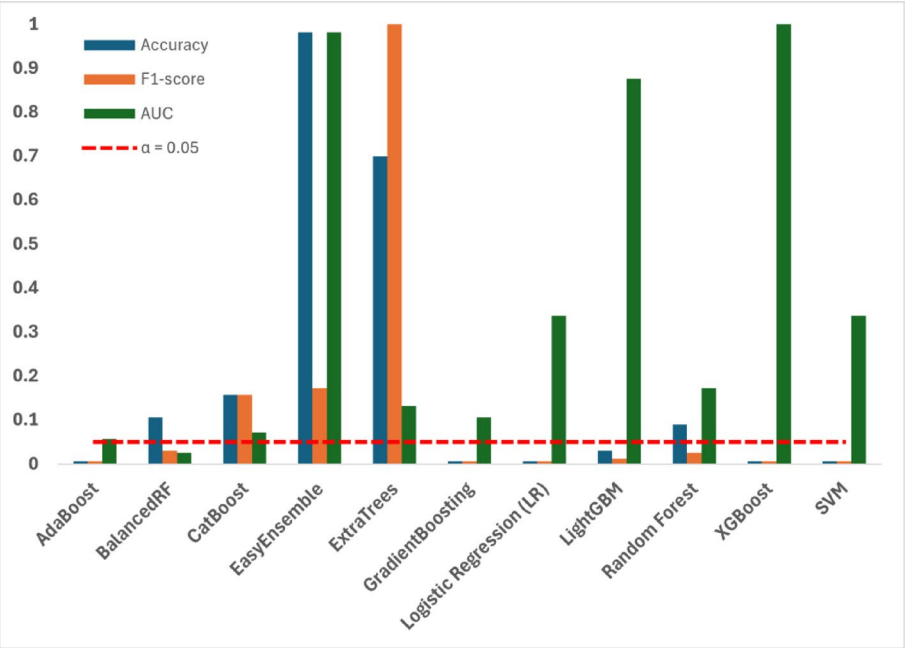


Fig. 5. FDR-adjusted *p*-values from Wilcoxon Signed-Rank tests comparing MCDELE against each baseline model across Accuracy, F1-score, and AUC. ($\alpha = 0.05$).

SHAP explanation for the base learners and meta-learner

In an attempt to enhance interpretability, we performed SHAP analysis by providing the individual contributions of features and ensemble-level meta-features to the predictions produced by MCDELE. These explanations describe the reasoning and feature utilization learned by the machine learning model, but should not be interpreted as reflecting causal relationships or mechanistic insight into disease. SHAP values were calculated for each tuned base learner to visualize interpretability at the base level, and global feature importance was obtained by taking each mean absolute SHAP value across all of the base models. For meta-level interpretability, SHAP was applied to the logistic regression (meta-learner) using base-model prediction probabilities, and diversity features were used as inputs.

Figure 6a represents the global feature importance, and Fig. 6b represents the local SHAP explanation for a representative patient correctly classified as having heart disease (True Class = 1, Predicted Class = 1). Positive SHAP values lead to higher estimated disease risk, and vice versa for negative SHAP values. Both the global and local SHAP feature importance show the importance of cluster-augmented features, cluster, and cluster_conf. Both features significantly improve the base learners' predictive ability. The 'cluster' feature shows higher importance than the clinical attributes like cholesterol, sex, and age, illustrating how well MCDELE uses the unsupervised structural patterns. Furthermore, 'cluster_conf' adds a new interpretability (Fig. 7).

In order to better interpret the clusters, we further evaluated their relationship with demographic and clinical variables. Chi-square tests showed that cluster assignments were significantly associated with heart disease status ($p = 0.001169$), age group ($p = 0.000002$), and sex ($p = 0.002975$). These results suggest that cluster-derived features capture population structure, with far-reaching implications on both demography and clinical characteristics of the dataset. Thus, the interpretation of the significance of cluster-based features should be viewed as a representation of data heterogeneity rather than as evidence for clinically relevant disease subtypes or causal disease mechanisms.

Conclusion

In this study, we have developed and evaluated a new stacking ensemble model called the Meta-Cluster Driven Ensemble Learning Engine (MCDELE) to predict heart disease. The proposed method is extensively tested on a publicly available Heart Failure dataset collected from Kaggle, which is derived from multiple UCI Heart Disease cohorts. The model leverages a two-level architecture. The base layer consists of a diverse range of base classifiers, whose prediction is enhanced through cluster-augmented features, cluster ID, and cluster confidence, obtained by using K-Means clustering. To improve the performance of the meta-learner, meta-features are introduced in the output vector of the base learners. MCDELE is thoroughly tested against a variety of powerful baseline and ensemble models. The key evaluation metrics include accuracy, precision, recall, F1-score, and AUC, and the Wilcoxon signed-rank test is used for statistical validation. The result clearly demonstrates that MCDELE:

- Attained the highest performance across several metrics, including accuracy (90.22), F1-score (91.26), precision (90.38), and a competitive AUC (93.77).
- Benefited from the cluster-augmented features, as confirmed by the SHAP summary plot.

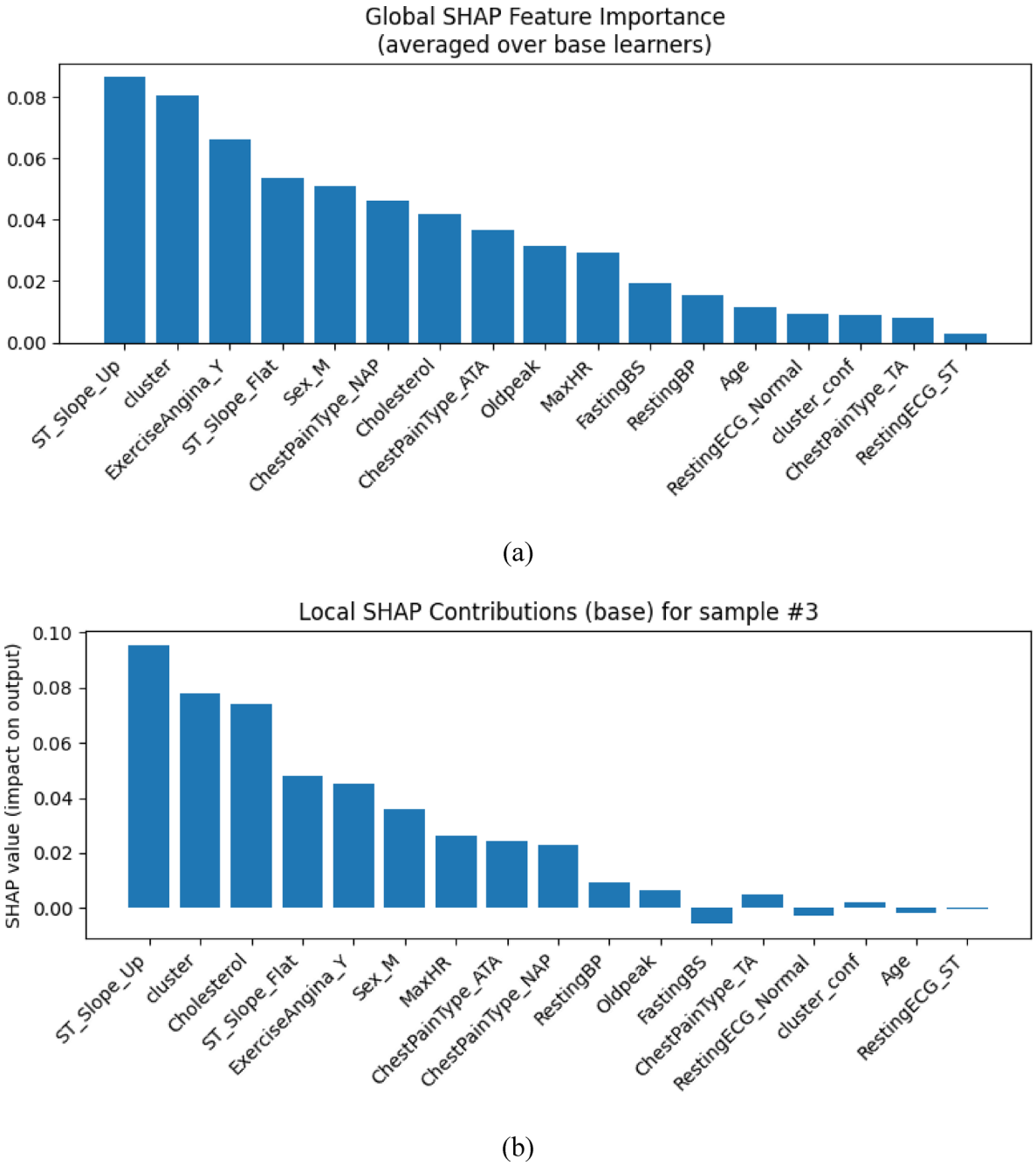
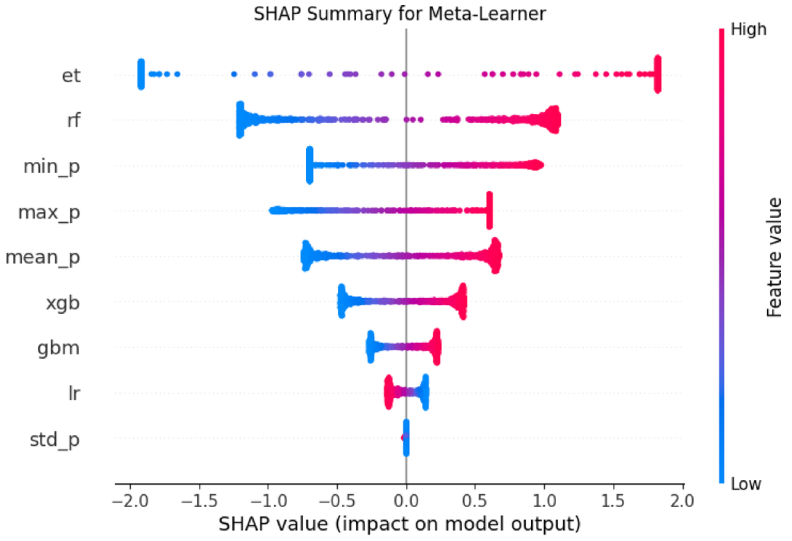


Fig. 6. (**a**) Global and (**b**) Local SHAP feature importance (Patient Example: True Positive) for the base learners.

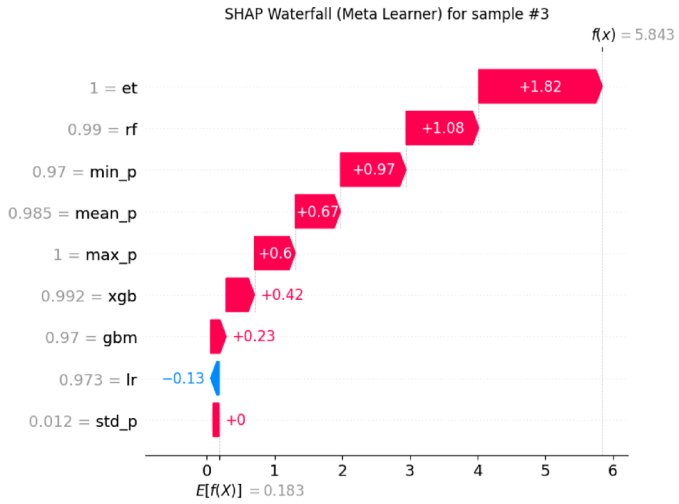
- Effectively utilized the proposed meta-features as evidenced by the SHAP summary and waterfall plot.

Furthermore, the Wilcoxon signed-rank test verified that MCDELE's improvements are statistically significant($p < 0.05$) over several competing methods across multiple matrices, while differences with a few strong ensemble models were not statistically significant for some matrices. The hybrid nature of the proposed ensemble combines the structure of the data by utilizing cluster features and predictive distribution via meta features. With feature-level insights made possible by SHAP, MCDELE is not only accurate but also interpretable.

Future work will focus on external clinical validation of the proposed framework, as well as established reporting and risk-of-bias assessment guidelines for AI-based clinical prediction, including TRIPOD + AI[53] and PROBAST + AI[54], to further improve the transparency, reproducibility, and clinical applicability.



(a)



(b)

Fig. 7. (**a**) Global and (**b**) Local SHAP feature importance (Patient Example: True Positive) for the meta-learners.

Data availability

The datasets analyzed during the current study are publicly available. The Heart Disease dataset from the UCI Machine Learning Repository is available at: https://archive.ics.uci.edu/dataset/45/heart+disease. The dataset provided by Kaggle (Lapp, D., Heart Disease Dataset, 2019) is also publicly accessible via the Kaggle platform. No new datasets were generated during the current study.

Code availability

The source code for the proposed MCDELE framework has been deposited in Zenodo (DOI: https://doi.org/10.5281/zenodo.21373512). The repository is currently under restricted access during the review process and will be made publicly available upon publication of this article.

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Author contributions

S.D. (Surajit Das) and S.P.N. (Samaleswari Prasad Nayak) contributed to the conceptualization and design of the study. B.S. (Biswajit Sahoo) and B.A. (Biswaranjan Acharya) were responsible for data curation, methodology, and implementation of the proposed model. S.C.R. (Satyananda Champati Rai) performed the formal analysis and validation. A.K.D. (Abdisa Kediro Dedefo) contributed to visualization, interpretation of results, and drafting of the manuscript.All authors contributed to reviewing and editing the manuscript and approved the final version.

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Declarations

Ethics approval and consent to participate

This study used the publicly available Heart Failure Prediction dataset available through Kaggle, which is derived from a de-identified and aggregated version of the UCI Heart Disease dataset. The original data providers anonymized all records prior to public release, and personally identifiable information is not included in the dataset. The authors did not collect new human-subject data, contact participants, or access identifiable patient information. Thus, this analysis is a secondary data analysis of publicly available de-identified data and does not require Institutional Review Board (IRB) approval and informed consent.

Competing interests

The authors declare no competing interests.

Additional information

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